Design and Construction of LPG Installations

API STANDARD 2510
SEVENTH EDITION, MAY 1995

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005



COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

Design and Construction of LPG Installations

Manufacturing, Distribution and Marketing Department

API STANDARD 2510
SEVENTH EDITION, MAY 1995

American
Petroleum
Institute



COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

SPECIAL NOTES

1. API PUBLICATIONS NECESSARILY ADDRESS PROBLEMS OF A GENERAL NATURE. WITH RESPECT TO PARTICULAR CIRCUMSTANCES, LOCAL, STATE, AND FEDERAL LAWS AND REGULATIONS SHOULD BE REVIEWED.

2. API IS NOT UNDERTAKING TO MEET THE DUTIES OF EMPLOYERS, MANUFACTURERS, OR SUPPLIERS TO WARN AND PROPERLY TRAIN AND EQUIP THEIR EMPLOYEES, AND OTHERS EXPOSED, CONCERNING HEALTH AND SAFETY RISKS AND PRECAUTIONS, NOR UNDERTAKING THEIR OBLIGATIONS UNDER LOCAL, STATE, OR FEDERAL LAWS.

3. INFORMATION CONCERNING SAFETY AND HEALTH RISKS AND PROPER PRECAUTIONS WITH RESPECT TO PARTICULAR MATERIALS AND CONDITIONS SHOULD BE OBTAINED FROM THE EMPLOYER, THE MANUFACTURER OR SUPPLIER OF THAT MATERIAL, OR THE MATERIAL SAFETY DATA SHEET.

4. NOTHING CONTAINED IN ANY API PUBLICATION IS TO BE CONSTRUED AS GRANTING ANY RIGHT, BY IMPLICATION OR OTHERWISE, FOR THE MANUFACTURE, SALE, OR USE OF ANY METHOD, APPARATUS, OR PRODUCT COVERED BY LETTERS PATENT. NEITHER SHOULD ANYTHING CONTAINED IN THE PUBLICATION BE CONSTRUED AS INSURING ANYONE AGAINST LIABILITY FOR INFRINGEMENT OF LETTERS PATENT.

5. GENERALLY, API STANDARDS ARE REVIEWED AND REVISED, REAFFIRMED, OR WITHDRAWN AT LEAST EVERY FIVE YEARS. SOMETIMES A ONE-TIME EXTENSION OF UP TO TWO YEARS WILL BE ADDED TO THIS REVIEW CYCLE. THIS PUBLICATION WILL NO LONGER BE IN EFFECT FIVE YEARS AFTER ITS PUBLICATION DATE AS AN OPERATIVE API STANDARD OR, WHERE AN EXTENSION HAS BEEN GRANTED, UPON REPUBLICATION. STATUS OF THE PUBLICATION CAN BE ASCERTAINED FROM THE API AUTHORING DEPARTMENT [TELEPHONE (202) 682-8000]. A CATALOG OF API PUBLICATIONS AND MATERIALS IS PUBLISHED ANNUALLY AND UPDATED QUARTERLY BY API, 1220 L STREET, N.W., WASHINGTON, D.C. 20005.

Copyright © 1995 American Petroleum Institute

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

FOREWORD

This standard provides minimum requirements for the design and construction of installations for the storage and handling of liquefied petroleum gas (LPG) at marine and pipeline terminals, natural gas processing plants, refineries, petrochemical plants, and tank farms. This standard takes into consideration the specialized training and experience of operating personnel in the type of installation discussed. In certain instances, exception to standard practices are noted and alternative methods are described.

This standard does not include information on the production or use of liquefied petroleum gas.

It is not intended that this standard be retroactive or that it take precedence over contractual agreements. Wherever practicable, existing codes and manuals have been used in the preparation of this standard.

This standard requires the purchaser to specify certain details and features. Although it is recognized that the purchaser may desire to modify, delete, or amplify sections of the standard, it is strongly recommended that such modifications, deletions, and amplifications be made by supplementing this standard rather than by rewriting or incorporating sections of this standard into another complete standard.

API standards are published as an aid to procurement of standardized equipment and materials. These standards are not intended to inhibit purchasers or producers from purchasing or producing products made to specifications other than those of API.

API publications may be used by anyone desiring to do so. Every effort has been made by the Institute to assure the accuracy and reliability of the data contained in them; however, the Institute makes no representation, warranty, or guarantee in connection with this publication and hereby expressly disclaims any liability or responsibility for loss or damage resulting from its use or for the violation of any federal, state, or municipal regulation with which this publication may conflict.

Suggested revisions are invited and should be submitted to the director of the Manufacturing, Distribution and Marketing Department, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

CONTENTS

		Page
SECTION 1—GENERAL		1
1.1	Scope	1
1.2	Retroactivity	1
1.3	Referenced Publications	1
1.4	Definitions	2
1.5	Characteristics of LPG	2
1.6	Safety	2
SECTION 2—DESIGN OF LPG TANKS		3
2.1	Applicable Design Construction Codes	3
2.2	Design Pressure and Temperature	3
2.3	Design Vacuum	3
2.4	Materials of Construction	3
2.5	Vessel Connections	3
2.6	Previously Constructed Vessels	3
SECTION 3—SITING REQUIREMENTS AND SPILL CONTAINMENT		4
3.1	Siting	5
3.2	Drainage	5
3.3	Spill Containment	5
3.4	Remote Impoundment	5
3.5	Diking	6
SECTION 4—FOUNDATIONS AND SUPPORTS FOR LPG STORAGE VESSELS AND RELATED PIPING		6
4.1	Applicable Codes and Specifications	6
4.2	Special Requirements	6
SECTION 5—TANK ACCESSORIES, INCLUDING PRESSURE- AND VACUUM-RELIEVING DEVICES		8
5.1	Mandatory Equipment	8
5.2	Tank Accessory Materials	10
SECTION 6—PIPING REQUIREMENTS		10
6.1	American Society of Mechanical Engineers Code for Pressure Piping	10
6.2	Pipe (Including Pressure Tubing and Excluding Instrument Piping)	10
6.3	Fittings	11
6.4	Plugs	11
6.5	Unions	11
6.6	Valves	11
6.7	Location, Installation, and Flexibility of Piping, Valves, and Fittings	11
SECTION 7—LOADING, PRODUCT TRANSFER, AND UNLOADING FACILITIES		12
7.1	Scope	12
7.2	Rates of Loading and Unloading	12
7.3	Transfer, Loading, and Unloading Equipment	12
7.4	Grounding and Bonding	13
7.5	Hose and Other Flexible Connectors for Product Transfer	13
7.6	Blowdown or Venting of Loading and Unloading Lines	14
7.7	Marking of Valves in Loading and Unloading Systems	14

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

7.8 Metering Equipment Used in Loading and Unloading 14
7.9 LPG Odorization 14

SECTION 8—FIRE PROTECTION 14
8.1 General 14
8.2 Access for Fire Fighting 14
8.3 Fire Water Use 14
8.4 Fire Detection Systems 16
8.5 Fire Extinguishers 16
8.6 Fire-Fighting Foam 16
8.7 Fireproofing of LPG Vessels 16
8.8 Fireproofing of Structural Supports 16
8.9 Burying and Mounding 17
8.10 Electrical Installations and Equipment 17
8.11 Critical Wiring and Control Systems 17
8.12 Safety Precaution Signs 17
8.13 Lighting 17
8.14 Fencing 17
8.15 Roadways 17

SECTION 9—REFRIGERATED STORAGE 18
9.1 General 18
9.2 Design Requirements 18
9.3 Siting Requirements 18
9.4 Thermal Considerations 19
9.5 Tank Accessories 19
9.6 Piping Requirements 20
9.7 Refrigeration System 20

APPENDIX A—PIPING, VALVES, FITTINGS, AND OPTIONAL EQUIPMENT .. 21

Table
1—Minimum Horizontal Distance Between Shell of Pressurized LPG Tank and Line of Adjoining Property That May Be Developed 4

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

Design and Construction of LPG Installations

SECTION 1—GENERAL

1.1 Scope

This standard covers the design, construction, and location of liquefied petroleum gas (LPG) installations at marine and pipeline terminals, natural gas processing plants, refineries, petrochemical plants, and tank farms. This standard covers storage vessels, loading and unloading systems, piping, or related equipment.

The size and type of the installation; the related facilities on the site; the commercial, industrial, and residential population density in the surrounding area; the terrain and climate conditions; and the type of LPG handled shall be considered. Generally speaking, the larger the installation and the greater the population density of the surrounding area, the more stringent are the design requirements.

Design and construction considerations peculiar to refrigerated storage, including autorefrigerated storage, are covered in Section 9 of this standard.

In this standard, numerical values are presented with U.S. customary units only. These U.S. customary values are to be regarded as the standard values.

This standard shall not apply to the design, construction, or relocation of frozen earth pits, underground storage caverns or wells, underground or mounded storage tanks, and aboveground concrete storage tanks.

This standard does not apply to the following installations:

a. Those covered by API Standard 2508.
b. Those covered by NFPA 58 and NFPA 59.
c. U.S. Department of Transportation (DOT) containers.
d. Gas utility company facilities; refinery equipment; gas processing equipment; and transfer systems from process equipment before LPG storage.
e. Those tanks with less than 2000 gallons of storage capacity.

1.2 Retroactivity

The provisions of this standard are intended for application to new installations. This standard can be used to review and evaluate existing storage facilities. However, the feasibility of applying this standard to facilities, equipment, structures, or installations that were already in place or that were in the process of construction or installation before the date of this publication, must be evaluated on a case-by-case basis considering individual circumstances and sites.

1.3 Referenced Publications

The most recent edition or revision of each of the following manuals, codes, recommended practices, publications, standards, and specifications shall form a part of this standard to the extent specified:

ACI[1]

318 *Building Code Requirements for Reinforced Concrete*

AISC[2]

Specification for Structural Steel Buildings

API

RP 500 *Classification of Locations for Electrical Installations at Petroleum Facilities*
RP 520 *Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries*
RP 521 *Guide for Pressure-Relieving and Depressuring Systems*
RP 550 *Manual on Installation of Refinery Instruments and Control Systems* (out of print)
RP 551 *Process Measurement Instrumentation*
Std 607 *Fire Test for Soft-Seated Quarter-Turn Valves*
Std 620 *Design and Construction of Large, Welded, Low-Pressure Storage Tanks*
RP 1102 *Steel Pipelines Crossing Railroads and Highways*
Std 2000 *Venting Atmospheric and Low-Pressure Storage Tanks: Nonrefrigerated and Refrigerated*
RP 2003 *Protection Against Ignitions Arising Out of Static, Lightning, and Stray Currents*
Publ 2218 *Fireproofing Practices in Petroleum and Petrochemical Processing Plants*
Std 2508 *Design and Construction of Ethane and Ethylene Installations at Marine and Pipeline Terminals, Natural Gas Processing Plants, Refineries, Petrochemical Plants, and Tank Farms*
Publ 2510A *Fire Protection Considerations for the Design and Operation of Liquefied Petroleum Gas (LPG) Storage Facilities*
Spec 6FA *Specification for Fire Test for Valves*

Manual of Petroleum Measurement Standards, Chapter 5, "Metering"

[1]American Concrete Institute, P. O. Box 19150, Detroit, Michigan 48219-0150.
[2]American Institute of Steel Construction, One East Wacker Drive, Suite 3100, Chicago, Illinois 60601-2001.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545849 883 ■

ASME[3]

B16.9 *Factory-Made Wrought Steel Buttwelding Fittings*

B31.3 *Chemical Plant and Petroleum Refinery Piping*

B31.4 *Liquid Transportation Systems for Hydrocarbons, Liquid Petroleum Gas, Anhydrous Ammonia, and Alcohols*

Boiler and Pressure Vessel Code, Section II, "Materials"; and Section VIII, "Pressure Vessels"

DOT[4]

Transportation Safety Act of 1974, Part 173, Section 315

ICBO[5]

Uniform Building Code

NFPA[6]

30 *Flammable and Combustible Liquids Code*

58 *Storage and Handling of Liquefied Petroleum Gases*

59 *Storage and Handling of Liquefied Petroleum Gases at Utility Gas Plants*

59A *Production, Storage and Handling of Liquefied Natural Gas (LNG)*

70 *National Electrical Code*

NPGA[7]

Bul 128 *Protection of Transfer Areas*

UL[8]

1709 *Rapid Rise Fire Tests of Protection Materials for Structural Steel*

1.4 Definitions

Some of the terms used in this standard are defined in 1.4.1 through 1.4.13.

1.4.1 An *aboveground tank* or *aboveground vessel* is a tank or vessel all or part of which is exposed above grade.

1.4.2 *Autorefrigeration* is the chilling effect of vaporization of LPG when it is released or vented to a lower pressure.

1.4.3 A *boiling-liquid expanding-vapor explosion* (*BLEVE*) is the phenomenon of a pressurized LPG tank failing such as can occur from direct exposure to a fire (normally a catastrophic event).

1.4.4 A *horizontal tank* or *horizontal vessel* is a pressure vessel used for the storage of LPG, installed with the vessel horizontal.

1.4.5 *Installations* are tanks, vessels, pumps, compressors, accessories, piping, and all other associated equipment required for the receipt, transfer, storage, and shipment of LPG.

1.4.6 *Liquefied petroleum gas* (*LPG* or *LP-gas*) is any material in liquid form that is composed predominantly of any of the following hydrocarbons or of a mixture thereof: propane, propylene, butanes (normal butane or isobutane), and butylenes.

1.4.7 A *mounded tank* or *mounded vessel* is a tank or vessel located above or partially above the general grade level but covered with earth, sand, or other suitable material.

1.4.8 *Refrigerated storage* is storage in a vessel or tank artificially maintained at a temperature below the nominal ambient temperature.

1.4.9 *Rollover* is the spontaneous and sudden movement of a large mass of liquid from the bottom to the top surface of a refrigerated storage reservoir due to an instability caused by an adverse density gradient. Rollover can cause a sudden pressure increase and can affect vessel integrity.

1.4.10 *Shall* indicates provisions that are mandatory.

1.4.11 Use of the term *shall consider* directly before a design or construction factor (such as a force or safety) indicates that the factor's effect and significance should be evaluated using good engineering judgement—through an examination or test if appropriate—and the design may or may not be adjusted accordingly.

1.4.12 A *tank* or *vessel* is a container used for storing LPG.

1.4.13 An *underground tank* or *underground vessel* is a tank or vessel all parts of which are completely buried below the general grade of the facility.

1.5 Characteristics of LPG

LPG is customarily handled in a liquid state achieved by its liquefaction under moderate pressure. Upon release of the pressure, LPG is readily converted into the gaseous phase at normal ambient temperature.

1.6 Safety

The safety of LPG storage installations is enhanced by the employment of good engineering practices, such as those recommended by this standard, during design and construction.

[3]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[4]U.S. Department of Transportation. The act is available from the U.S. Government Printing Office, Washington, D.C. 20402.
[5]International Conference of Building Officials, 5360 Workman Mill Road, Whittier, California 90601-2298.
[6]National Fire Protection Association, 1 Batterymarch Park, Quincy, Massachusetts 02269-9101.
[7]National Propane Gas Association, 1600 Eisenhower Lane, Suite 100, Lisle, Illinois 60532.
[8]Underwriters Laboratories, 333 Pfingsten Road, Northbrook, Illinois 60062-2096.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545850 5T5 ■

SECTION 2—DESIGN OF LPG TANKS

2.1 Applicable Design Construction Codes

2.1.1 Vessels fabricated after the publication of this standard shall meet the requirements of the ASME *Boiler and Pressure Vessel Code*, Section VIII, Division 1 or 2.

2.1.2 When complete rules for any specific design are not given, the manufacturer, subject to the approval of the purchaser, shall provide a design as safe as would be provided in the currently applicable code listed in 2.1.1.

2.2 Design Pressure and Temperature

2.2.1 The design pressure of LPG vessels shall not be less than the vapor pressure of the stored product at the maximum product design temperature. The additional pressure resulting from the partial pressure of noncondensable gases in the vapor space and the hydrostatic head of the product at maximum fill shall be considered. Ordinarily, the latter considerations and the performance specifications of the relief valve require a differential between design pressure and maximum product vapor pressure that is adequate to allow blowdown of the pressure relief valve (see API Recommended Practice 520).

2.2.2 Both a minimum design temperature and a maximum design temperature shall be specified. In determining a maximum design temperature, consideration shall be given to factors such as ambient temperature, solar input, and product run down temperature. In determining a minimum design temperature, consideration shall be given to the factors noted in the preceding sentence as well as the autorefrigeration temperature of the stored product when it flashes to atmospheric pressure.

2.3 Design Vacuum

LPG storage vessel design shall consider vacuum effects and be designed accordingly. Where an LPG vessel is not designed for full vacuum, some alternatives, in order of preference, are as follows:

a. *Design for partial vacuum with a vacuum relief valve and a connection to a reliable supply of inert gas.* This alternative requires a means of venting inert gas from the storage vessel after it is no longer required for maintaining pressure.

b. *Design for partial vacuum with a vacuum relief valve and a connection to a reliable supply of hydrocarbon gas.* This alternative may compromise product quality.

c. *Design for partial vacuum with a vacuum relief valve that admits air to the vessel.* This alternative, under some conditions, may present a hazard from the presence of air in the LPG storage vessel, and this hazard shall be considered in the design.

2.4 Materials of Construction

2.4.1 All materials of construction shall meet the requirements of Section II of the ASME *Boiler and Pressure Vessel Code*.

2.4.2 Low-melting-point materials of construction, such as aluminum and brass, shall not be used for LPG storage vessels.

2.5 Vessel Connections

2.5.1 Shell penetrations in the bottom of the vessel between vessel supports should be avoided in the design of horizontal cylindrical vessels used to store LPG. The number of penetrations in any storage vessel shall be minimized.

2.5.2 Flange connections shall be a minimum of ASME 150-pound flange class. All fittings shall be a minimum of ¾ NPS.

2.5.3 Refer to Section 6 for piping requirements.

2.6 Previously Constructed Vessels

Vessels constructed prior to the issuance of this standard may be repaired or maintained and continued in service under the following conditions:

a. Operating conditions shall be the same as or less severe than those under which they are currently operating.
b. Regular inspections are performed.
c. Maximum allowable pressures are downgraded when inspection indicates the downgrade is appropriate.

Where an existing vessel is to be reused in a newly constructed facility, the code to which the vessel was originally designed is acceptable, provided that the vessel is inspected and verified as suitable for the intended new service.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545851 431 ■

SECTION 3—SITING REQUIREMENTS AND SPILL CONTAINMENT

3.1 Siting

3.1.1 GENERAL

3.1.1.1 Site selection is meant to minimize the potential risk to adjacent property presented by the storage facility and the risk presented to the storage facility by a fire or explosion on adjacent property. The following factors shall be considered in site selection:

a. Proximity to populated areas.
b. Proximity to public ways.
c. Risk from adjacent facilities.
d. Storage quantities.
e. Present and predicted development of adjacent properties.
f. Topography of the site, including elevation and slope.
g. Access for emergency response.
h. Availability of needed utilities.
i. Requirements for the receipt and shipment of products.
j. Local codes and regulations.

3.1.1.2 With the exception of spacing, the design features discussed in this standard are intended to prevent a major incident. Spacing is intended to minimize both the potential for small leak ignition and the exposure risk presented to adjacent vessels, equipment, or installations in case ignition occurs. Spacing is not intended to provide protection from a major incident.

3.1.1.3 Safety analysis and dispersion modeling are useful tools in estimating setback distances to limit the exposure risk to adjacent facilities.

3.1.2 MINIMUM DISTANCE REQUIREMENT

3.1.2.1 The minimum horizontal distance between the shell of a pressurized LPG tank and the line of adjoining property that may be developed shall be as shown in Table 1.

Where residences, public buildings, places of assembly, or industrial sites are located on adjacent property, greater distances or other supplemental protection shall be provided.

3.1.2.2 The minimum horizontal distance between the shells of pressurized LPG tanks or between the shell of a pressurized LPG tank and the shell of any other pressurized hazardous or flammable storage tank shall be as follows:

a. Between two spheres, between two vertical vessels, or between a sphere and a vertical vessel, 5 feet or half of the diameter of the larger vessel, whichever is greater.
b. Between two horizontal vessels, or between a horizontal vessel and a sphere or vertical vessel, 5 feet or three quarters of the diameter of the larger vessel, whichever is greater.

3.1.2.3 The minimum horizontal distance between the shell of a pressurized LPG tank and the shell of any other nonpressurized hazardous or flammable storage tank shall be the largest of the following with the exception noted after Item d:

a. If the other storage is refrigerated, three quarters of the greater diameter.
b. If the other storage is in atmospheric tanks and is designed to contain material with a flash point of 100°F or less, one diameter of the larger tank.
c. If the other storage is in atmospheric tanks and is designed to contain material with a flash point greater than 100°F, half the diameter of the larger tank.
d. 100 feet.

The minimum horizontal distance between shells need not exceed 200 feet.

3.1.2.4 The minimum horizontal distance between the shell of an LPG tank and a regularly occupied building shall be as follows:

a. If the building is used for the control of the storage facility, 50 feet.
b. If the building is used solely for other purposes (unrelated to control of the storage facility), 100 feet.

3.1.2.5 The minimum horizontal distance between the shell of an LPG tank and facilities or equipment not covered in 3.1.2.1 through 3.1.2.4 shall be as follows:

a. For process vessels, 50 feet.
b. For flares or other equipment containing exposed flames, 100 feet.
c. For other fired equipment, including process furnaces and utility boilers, 50 feet.
d. For rotating equipment, 50 feet; except for pumps taking suction from the LPG tanks, 10 feet.
e. For overhead power transmission lines and electric substations, 50 feet. In addition, siting shall be such that a

Table 1—Minimum Horizontal Distance Between Shell of Pressurized LPG Tank and Line of Adjoining Property That May Be Developed

Water Capacity of Each Tank (gallons)	Minimum Distance (feet)
2,000–30,000	50
30,001–70,000	75
70,001–90,000	100
90,001–120,000	125
120,001 or greater	200

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545852 378 ■

break in the overhead lines shall not cause the exposed ends to fall on any vessel or equipment.
f. For loading and unloading facilities for trucks and railcars, 50 feet.
g. For navigable waterways, docks, and piers, 100 feet.
h. For stationary internal combustion engines, 50 feet.

3.1.2.6 The minimum horizontal distance between the shell of an LPG tank and the edge of a spill containment area for flammable or combustible liquid storage tanks shall be 10 feet.

Note: If the spill containment is by the use of dikes or walls, the edge of the spill containment area for the purpose of spacing is defined as the centerline of the dike or wall. If the spill containment is by sloping, grading, or channels, the edge of the spill containment area for the purpose of spacing is defined as the outer edge of the wetted area at the design incident for the spill containment facility.

3.1.3 SITING OF PRESSURIZED LPG TANKS AND EQUIPMENT

3.1.3.1 Pressurized LPG tanks shall not be located within buildings, within the spill containment area of flammable or combustible liquid storage tanks as defined in NFPA 30, or within the spill containment area for refrigerated storage tanks.

3.1.3.2 Compressors and pumps taking suction from the LPG tanks shall not be located within the spill containment area of any storage facility. As an exception, either of the following is permitted to be used: (a) a submerged-motor, direct-coupled pump with no rotating element outside of the pump containment vessel; (b) a submersible pump inside an LPG tank.

3.1.3.3 Horizontal LPG tanks with capacities of 12,000 gallons or greater shall not be formed into groups of more than six tanks each. Where multiple groups of horizontal LPG vessels are to be provided, each group shall be separated from adjacent groups by a minimum horizontal shell-to-shell distance of 50 feet.

Note: Horizontal vessels used to store LPG should be oriented so that their longitudinal axes do not point toward other containers, process equipment, control rooms, loading or unloading facilities, or flammable or combustible liquid storage facilities located in the vicinity of the horizontal vessel.

3.2 Drainage

3.2.1 The ground under and surrounding a vessel used to store LPG shall be graded to drain any liquid spills to a safe area away from the vessel and piping. The grading shall be at a slope of at least 1 percent.

3.2.2 The drainage system shall be designed to prevent liquid spilled from one tank from flowing under any other tank and shall minimize the risk to piping from spilled LPG.

3.2.3 The spill drainage area shall not contain equipment, except as permitted by this standard.

3.2.4 Walls, dikes, trenches, or channels are permitted to assist in draining the area.

3.3 Spill Containment

3.3.1 Spill containment shall be considered for all locations and provided in locations in which either of the following conditions will result in a significant hazard:

a. The physical properties of the stored LPG make it likely that liquid LPG will collect on the ground. (This would be the case if the LPG is a mixture of butane and pentane.)
b. Climatic conditions during portions of the year make it likely that liquid LPG will collect on the ground.

3.3.2 The following shall be considered in the selection of materials for all components—including structural supports—of a spill containment facility:

a. The effects of thermal shock associated with spilling LPG (such as shock resulting from the autorefrigeration temperature).
b. Provision of adequate venting of the vapor generated during an LPG spill.

3.3.3 If spill containment is to be provided, it shall be by remote impoundment of spilled material or by diking of the area surrounding the vessel. The containment area shall not contain any other equipment, except as permitted by this standard.

3.4 Remote Impoundment

3.4.1 If remote impoundment is to be used for spill containment, the remote impoundment facility shall be designed according to the requirements given in 3.4.2 through 3.4.5.

3.4.2 Grading of the area under and surrounding the vessels shall direct any liquid leaks or spills to the remote impoundment area. Grading shall be at a minimum of 1 percent slope.

3.4.3 The use of walls, dikes, trenches, or channels to facilitate the draining of the area is permitted.

3.4.4 The remote impoundment area shall be located at least 50 feet from the vessels draining to it and from any hydrocarbon piping or other equipment.

3.4.5 The holdup of the remote impoundment area shall be at least 25 percent of the volume of the largest vessel draining to it. If the material stored in the vessel has a vapor pressure that is less than 100 pounds per square inch absolute at 100°F, the holdup for the remote impoundment facility shall be at least 50 percent of the volume of the largest vessel draining to it. Larger holdups shall be provided in the remote impoundment facility at locations where the expected vaporization is less than that indicated by the material's vapor

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

pressure because of climatic conditions or the physical properties of the material.

3.5 Diking

3.5.1 If diking around the vessel is to be used for spill containment, the diked area shall be designed according to the requirements given in 3.5.2 through 3.5.7.

3.5.2 Grading of the area under and surrounding the vessels shall direct any liquid leaks or spills to the edge of the diked area. Grading shall be at a minimum of 1 percent slope. Within the diked area, grading should cause spills to accumulate away from the vessel and any piping located within the diked area.

3.5.3 If an LPG sphere is diked, each sphere shall be provided with its own diked area. If LPG is stored in horizontal vessels, a single diked area may serve a group of tanks, as defined in 3.1.3.3.

3.5.4 The holdup of the diked area shall be at least 25 percent of the volume of the largest vessel within it. If the material stored in the vessel has a vapor pressure that is less than 100 pounds per square inch absolute at 100°F, the holdup for the diked area shall be at least 50 percent of the volume of the largest vessel within it. Larger holdups shall be provided in the diked area at locations where the expected vaporization is less than that indicated by the material's vapor pressure because of climatic conditions or the physical properties of the material.

Note: Larger holdups may also be provided when more than one vessel is located within the same diked area.

3.5.5 When dikes or walls are used as part of the spill containment system, the minimum height of a dike or wall constructed of earth shall be 1.5 feet and the minimum height of a dike or wall constructed of concrete, masonry, or another erosion-resistant material shall be 1 foot. Where dikes must be higher than 6 feet, provisions shall be made for normal and emergency access into and out of the diked enclosure. Where dikes must be higher than 12 feet or where ventilation is restricted by the dike, provision shall be made for normal operation of valves and access to the top of the tank or tanks without the need for personnel to enter into the area of the diked enclosure that is below the top of the dike. All earthen dikes shall have a flat top section not less than 2 feet wide.

3.5.6 Any dike or wall enclosure used for LPG containment shall include adequate access provisions (such as stairs for personnel and ramps for vehicles, if required), shall be designed to permit its free ventilation, and shall be constructed to retain the spilled liquid. Enclosures shall be designed to prevent unauthorized access by motor vehicles.

3.5.7 If the floor of any dike or wall enclosure used for LPG containment will not allow rainwater to dissipate within 24 hours, a drainage system shall be installed. Any drainage system provided shall include a valve or shear gate located in an accessible position outside the enclosure. The valve or shear gate shall normally be kept closed. The drainage system shall be one of the following types:

a. A vapor sealed catch basin within the enclosure discharging to a closed drainage system outside the enclosure.
b. A pipe through the dike or wall discharging to a drainage system outside the enclosure.

The drainage system shall keep the contents of the tank from entering natural water courses and from entering systems incapable of safely containing LPG.

SECTION 4—FOUNDATIONS AND SUPPORTS FOR LPG STORAGE VESSELS AND RELATED PIPING

4.1 Applicable Codes and Specifications

The materials, principles, methods, and details of design and construction of foundations and supports for LPG storage vessels and related piping shall meet the requirements stipulated in the following codes and specifications:

a. For concrete, ACI 318.
b. For masonry, ICBO *Uniform Building Code.*
c. For structural steel, AISC *Specification for Structural Steel Buildings.*

Where applicable local codes are more stringent, the local codes shall apply.

4.2 Special Requirements

4.2.1 GENERAL

The foundation and supports shall conform to the provisions set forth in 4.2.2 through 4.2.15.

4.2.2 MATERIALS

Supporting structures shall be made of one or a combination of the following materials:

a. Reinforced masonry.
b. Reinforced concrete.
c. Steel plate, pipe, or structural shapes.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

4.2.3 SOIL INFORMATION

The design of the foundation shall be based on a thorough knowledge of the load-bearing capacity and settlement properties of the soil. Where information regarding soil conditions is not available, an investigation shall be conducted.

4.2.4 SETTLEMENT OF FOUNDATION

The size and depth of the foundation shall be designed to limit settlement of the vessel to prevent excessive stresses in the tank and connected piping.

Note: Settlement should be monitored with regard to extent and uniformity, distribution of loads, changes in weather conditions (such as freezing and thawing, rain, or temporary drying out), types of soil, movement of adjoining foundations, and shifting of adjacent soil.

4.2.5 BOTTOM OF FOUNDATION

The bottom of the foundation shall be below the frost line and below nearby sewers or lines having the potential for leakage or washout that could result in settlement of the foundation.

4.2.6 FLOATING FOUNDATION OR PILING

Where it is impracticable to design foundations for normal settlement as described in 4.2.4, a floating foundation or piling is permitted. In this case, the settlement indicated by soil tests shall be used for design, and the settlement measured during subsequent service shall be used to check for adequate flexibility in connected piping.

4.2.7 LOADS ON SUPPORTING STRUCTURE

The following loads shall be considered in the design of the supporting structure:

a. Static loads during erection plus expected wind, ice, and snow loads during the erection.
b. Static loads during water testing plus 25 percent of the wind, ice, and snow loads.
c. Static loads during operation (including the load due to fireproofing) plus wind, ice, snow, and earthquake loads where required by local conditions.
d. Loads resulting from expansion and contraction of the vessel due to internal pressure and temperature changes.
e. Loads resulting from differential settlement across the supporting structures and foundations.
f. Static and dynamic loads during maintenance and operations.

4.2.8 SUPPORT DESIGN

4.2.8.1 The design of supports for vessels shall include provisions for expansion and contraction of the vessel due to internal pressure and temperature change of the vessel shell.

4.2.8.2 Flexibility shall be provided in the attached piping to avoid imposing excessive stress on vessel nozzles and associated piping as a result of vessel movement.

Note: The following publications contain additional material regarding the design of supports:

a. Section VIII of the ASME *Boiler and Pressure Vessel Code*.
b. L. P. Zick's article.[9]
c. *Pressure Vessels and Piping: Design and Analysis*,[10] which refers to a method of L. P. Zick.

4.2.9 VESSEL SHELL LOADS

In the design of vessel supports, special attention shall be given to the loads imposed on the vessel shell. Consideration shall be given to the following:

a. Secondary forces resulting from service temperatures or changes in temperatures.
b. Test and operating pressures.
c. Liquid loads, both with and without pressure applied.
d. Loads due to piping reactions.
e. Normal supporting loads.
f. Loads due to liquid sloshing (in earthquake zones).

4.2.10 DIAGONAL MEMBERS

Diagonal members, such as those used for bracing vertical columns, shall not be attached directly to a vessel unless adequate provisions are made for the resulting loads in the design of the vessel.

4.2.11 SADDLES

4.2.11.1 When a horizontal tank is supported by saddles, the features specified in 4.2.11.2 through 4.2.11.5 shall be incorporated in the design.

4.2.11.2 Two piers shall be used to support horizontal vessels.

4.2.11.3 Consideration shall be given to the placement of supports to obtain the most desirable stress distribution in the vessel shell.

4.2.11.4 The shape of the saddles shall conform to the fabricated shape of the vessel or to the steel pad attached to the vessel.

4.2.11.5 Doublers or reinforcing plates may be installed between the vessel shell and the supports to avoid external corrosion of the shell, provide for wear caused by temperature-induced movement, or reduce the stress in the shell at the support points. If such plates are used, they shall be

[9]L. P. Zick, "Stresses in Large Horizontal Cylindrical Pressure Vessels on Two Saddle Supports," *Welding Research Council Research Reports*, supplement to *Welding Journal*, Volume 30, No. 9, September 1951.
[10]*Pressure Vessels and Piping: Design and Analysis*, Volume II, ASME, Fairfield, NJ, 1972.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545855 087 ■

continuously welded to the vessel shell after any free moisture is removed from under the plates. A threaded weep hole shall be provided at the low point of each plate. Where corrosion plates are used, the plates shall extend beyond the limits of the supporting saddles to aid in distributing the support loads. The thickness of corrosion plates shall not be included in calculating the stress at the horn of the saddle.

4.2.12 MULTIPLE VESSELS

4.2.12.1 Continuous footings may be used for multiple vessel installations. In such instances, the loading of footings shall be calculated for various probable combinations of loads, such as the load that occurs when adjacent vessels are full and the load that occurs when alternate vessels are full.

4.2.12.2 Continuous piers shall not be used for multiple vessel installations without the incorporation of special drainage provisions.

4.2.13 ANCHORAGE

4.2.13.1 In areas where there is a risk of flooding, the vessel shall be anchored to the foundation or support to prevent floating in case of a flood. Anchorage shall not restrict vessel movements resulting from expansion and contraction of the vessel due to temperature changes and internal pressure.

4.2.13.2 Anchorage of the vessel to the foundation or support shall be provided to resist wind and earthquake loads and to control temperature-induced movement.

4.2.13.3 Anchorage to the foundation or support shall be provided to resist any uplifting forces resulting from internal pressure in the tank or vessel.

4.2.14 VERTICAL TANK SKIRTS

4.2.14.1 Where vertical vessels are supported by skirts, the skirts shall be provided with a single opening for inspection or access. The opening shall be as small as practicable.

4.2.14.2 Skirt openings shall be reinforced when required to prevent buckling or overstressing of the skirt as a result of imposed loads as covered in 4.2.7.

4.2.15 CORROSION PROTECTION

4.2.15.1 Steel supports and their members shall be positioned to prevent the accumulation of water. Where this positioning is impractical, adequate drainage openings shall be provided to prevent such accumulation.

4.2.15.2 Enclosed spaces in which water might accumulate during construction or operation shall be provided with drainage openings.

SECTION 5—TANK ACCESSORIES, INCLUDING PRESSURE- AND VACUUM-RELIEVING DEVICES

5.1 Mandatory Equipment

5.1.1 GENERAL

Tanks shall be fitted with the equipment described in 5.1.2 through 5.1.8. Equipment shall be suitable for use with LPG and designed for at least the maximum service conditions to which it may be subjected.

5.1.2 LIQUID-LEVEL GAUGING EQUIPMENT

5.1.2.1 Each LPG tank shall be provided with liquid-level gauging equipment as specified in 5.1.2.2 through 5.1.2.5.

5.1.2.2 Each tank shall be equipped with a reliable level-indicating system. The need for a second, independent level-indicating system shall be determined by a safety analysis.

5.1.2.3 An independent high-level alarm shall be provided. The alarm shall be set to give the operator sufficient time to stop the flow before the maximum permissible filling height is exceeded (see 5.1.3). The alarm shall be located so that it is audible and visible to the operating personnel controlling the filling operation.

5.1.2.4 For tanks that cannot be removed from service, provisions shall be included for testing, repairing, and replacing primary gauges and alarms while the tank is in service.

5.1.2.5 In tanks that have a high-level cutoff, the cutoff device shall be in addition to and independent of the high-level alarm specified in 5.1.2.3.

5.1.3 MAXIMUM LIQUID LEVEL

The maximum permissible filling height of an LPG tank shall be set to provide adequate vapor space to accommodate any thermal expansion that may occur after filling is completed. The maximum filling height shall be set so that when a tank filled to that level at the minimum anticipated storage temperature is exposed to a fire, the thermal expansion of the liquid will not cause the LPG level to exceed 98 percent of the liquid full level at the set pressure of the pressure relief valve.

5.1.4 LEVEL GAUGES

Columnar glass level gauges shall not be used. Reflex and see-through level gauges shall be equipped with a ball check valve or a similar protective device.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545856 T13 ■

5.1.5 PRESSURE GAUGE

On each tank, a suitable pressure gauge shall be connected to the vapor space.

5.1.6 PRESSURE- AND VACUUM-RELIEVING DEVICES

5.1.6.1 General

Each tank shall be provided with one or more spring-loaded or pilot-operated pressure relief valves. The pressure relief valve or valves shall be set to discharge as required by the ASME Code. Pilot-operated pressure relief devices shall be designed so that the main valve will open automatically and protect the tank if the pilot valve fails. Pilot-operated valves shall be provided with a backflow preventer. Tanks that may be damaged by internal vacuum shall be provided with vacuum-relieving devices. Weight and lever pressure-relieving devices shall not be used.

5.1.6.2 Pressure Relief Valve Flow Capacities

Pressure relief valves installed on LPG tanks shall be designed to provide adequate flow capacity to protect the tank during fire exposure. Other causes of tank overpressure, such as overfilling and introduction of material with a higher vapor pressure in a common piping system, shall be considered in determining design flow capacity. Pressure relief valves shall be designed and sized in accordance with API Recommended Practice 520, Part I, and API Recommended Practice 521.

5.1.6.3 Pressure Relief Valve Information

Each pressure relief valve shall be marked as required by the applicable ASME code, API standard, or API recommended practice.

5.1.6.4 Pressure Relief Valve Installation

5.1.6.4.1 Pressure relief valves shall be installed in accordance with API Recommended Practice 520, API Recommended Practice 521, and the requirements of 5.1.6.4.2 through 5.1.6.4.6.

5.1.6.4.2 The relief valve shall be installed to provide direct connection to the vapor space and to minimize liquid carry-over during vapor relief, especially when the tank is nearly full. This shall be achieved by locating relief valve connections at the top of the vapor space.

5.1.6.4.3 The possibility of tampering with the adjustment mechanism shall be minimized. If the adjustment mechanism is external, it shall be sealed.

5.1.6.4.4 The inlet and outlet piping for the pressure relief valve shall be designed to pass the required flow without exceeding the allowable pressure-drop limits.

5.1.6.4.5 The pressure relief system shall be protected from the closure of any block valves installed between the tank and the pressure relief valve or between the pressure relief valve and its discharge vent outlet. This protection may be achieved by one of the following procedures:

a. *Installing the pressure relief valve without block valves.*
b. *Providing excess pressure relief valve capacity with multiway valves, interlocked valves, or sealed block valves arranged so that isolating one pressure relief valve will not reduce the capacity of the system to below the required relieving capacity.*
c. *Locking or sealing the block valves open without installing excess relieving capacity, as follows.* The valves shall be closed by an authorized person who shall remain at the valves while they are closed and the tank is in operation and shall lock or seal the valves open before leaving. The authorized person shall be able to observe the operating pressure while the valves remain blocked and shall be ready to take emergency action if required.

5.1.6.4.6 The stem of any gate valve installed in the pressure relief system shall be in a horizontal or below-centerline position.

5.1.6.5 Discharge Vents

5.1.6.5.1 Discharge vents from the pressure relief valves or common discharge headers shall be designed to meet the requirements of API Recommended Practice 520 and API Recommended Practice 521 and shall be installed in accordance with the requirements given in 5.1.6.5.2 through 5.1.6.5.6.

5.1.6.5.2 Discharge vents shall lead to the open air or to a flare system. Discharging directly to the atmosphere is unacceptable if liquid LPG might be released into the atmosphere, unless the discharge is through thermal relief valves. Positive design and operational steps shall be taken to prevent the discharge of liquid LPG from atmospheric vents. Such steps include automatic shutdown of filling operations prior to overfilling.

5.1.6.5.3 Discharge vents shall be protected against mechanical damage.

5.1.6.5.4 If discharge vents relieve to the atmosphere, they shall be designed to prevent entry of moisture and condensate. This design may be accomplished by the use of loose-fitting rain caps and drains. Drains shall be installed so that the discharge will not impinge on the tank or adjoining tanks, piping, equipment, and other structures.

5.1.6.5.5 Discharge vents shall be designed to handle any thrust developed during venting.

5.1.6.5.6 Discharge shall be to an area that has the following characteristics:

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545857 95T ■

a. The area prevents flame impingement on tanks, piping, equipment, and other structures.
b. The area prevents vapor entry into enclosed spaces.
c. The area is above the heads of any personnel on the tank, adjacent tanks, stairs, platforms, or the ground.

5.1.6.6 Pressure Setting

Pressure relief valves shall be tested for correct set pressure before being placed in service. See API Recommended Practice 520.

5.1.7 SHUTOFF VALVES

5.1.7.1 Shutoff valves shall conform to the criteria specified in 5.1.7.1.1 through 5.1.7.1.3.

5.1.7.1.1 Shutoff valves shall be provided for all tank connections except the following:

a. Connections on which safety valves are mounted.
b. Connections containing a ⅛-inch-maximum restriction orifice, plugs, or thermometer wells.

5.1.7.1.2 Shutoff valves shall be located as close to the tank as is practical. The preferred location is at the shell nozzle. Shutoff valves shall be readily accessible for operation and maintenance. As an alternative, shutoff valves are permitted to be located on vessel nozzles that extend beyond the spill path or containment area, provided the extended nozzles are suitably protected from freezing and fire exposure.

5.1.7.1.3 Shutoff valves shall conform to the material and construction requirements of 6.6.

5.1.7.2 All shutoff valves located on nozzles below the maximum liquid level shall be designed to provide a visual indication of the valve position and shall be capable of maintaining an adequate seal under fire conditions. Valves meeting the requirements of API Standard 607 or API Specification 6FA have the required fire resistance.

5.1.7.3 When the capacity of the vessel exceeds 10,000 gallons, all shutoff valves on inlet and outlet piping located below the maximum liquid level shall either close automatically or be remotely operable during the first 15 minutes of fire exposure. This may require fireproofing of the control system (see 8. 11). These valves shall also be manually operable at the installed location. Check valves installed on dedicated fill lines are suitable for meeting the requirements of this paragraph.

5.1.8 TEMPERATURE INDICATOR

Each tank shall be fitted with a suitable thermometer well.

5.2 Tank Accessory Materials

Ductile (nodular) iron, cast aluminum, malleable iron, and brass shall not be used in any pressure-retaining tank accessories.

SECTION 6—PIPING REQUIREMENTS

6.1 American Society of Mechanical Engineers Code for Pressure Piping

Piping at facilities covered under this standard shall conform to the provisions of ASME B31.3; except that piping that falls under the exclusion provided in 300.1.3(e) of ASME B31.3 shall be constructed in accordance with the provisions of ASME B31.4. The additional provisions of this section apply to piping constructed in accordance with ASME B31.3.

6.2 Pipe (Including Pressure Tubing and Excluding Instrument Piping)

6.2.1 RECOMMENDED PIPE

Piping shall be seamless, electric-resistance-welded, or submerged-arc-welded pipe. Pipe to be used in piping applications of 2 inches or smaller shall be seamless.

6.2.2 PIPING JOINTS

6.2.2.1 The number of joints of any type between the vessel and the first block valve shall be minimized.

6.2.2.2 Welded joints shall be used where practical.

6.2.2.3 The number of flanged or threaded joints shall be minimized.

6.2.2.4 Joints in pipe 2 NPS or larger shall be welded or flanged.

6.2.2.5 Joints in pipe smaller than 2 NPS shall be welded, threaded, or flanged.

6.2.2.6 Piping gaskets shall be of the self-centering or confined type and shall be resistant to LPG.

6.2.2.7 Threaded connections between the vessel and the first valve where flow can be shut off shall be made up dry and seal welded. The seal weld shall cover all exposed threads. A male-female valve screwed directly into a fitting on the vessel is permitted in lieu of seal welding.

6.2.3 MINIMUM SPECIFICATIONS

6.2.3.1 The pipe wall thickness shall be equal to or greater than that required by ASME B31.3. The minimum requirements specified in 6.2.3.2 and 6.2.3.3 shall also apply.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545858 896 ■

6.2.3.2 Pipes made from notch-sensitive materials, such as carbon steel, shall have the following minimum wall thicknesses:

a. Nominal pipe size less than 2 NPS-Schedule 80.
b. Nominal pipe size of 2-5 NPS-Schedule 40 (except for threaded connections, which shall be Schedule 80).
c. Nominal pipe size of 6 NPS-wall thickness of 0.25 NPS.
d. Nominal pipe size of 8-12 NPS-Schedule 20.
e. Nominal pipe size of 14 NPS or larger-Schedule 10.

6.2.3.3 Pipes made from non-notch-sensitive material, such as stainless steel, shall have the following minimum wall thicknesses:

a. Nominal pipe size of ¾ NPS or less-Schedule 80S.
b. Nominal pipe size of 1, 1½, or 2 NPS-Schedule 40S.
c. Nominal pipe size larger than 2 NPS-Schedule 10S.

6.2.4 PRESSURE TUBING

Tubing shall be constructed of steel. If tubing will be exposed to a corrosive atmosphere, stainless steel shall be used.

6.3 Fittings

6.3.1 THREADED FITTINGS AND COUPLINGS

Threaded fittings 1½ NPS or smaller in size, such as elbows, tees, and crosses, shall be of forged steel and shall have a working pressure of at least 3000 pounds per square inch. Threaded couplings shall be of forged steel and shall have a working pressure of at least 3000 pounds per square inch.

6.3.2 BUTT-WELDING FITTINGS

Butt-welding fittings shall be made from seamless steel or equivalent material, shall be of at least the same thickness and schedule as the piping, and shall conform to ASME B16.9.

6.3.3 SOCKET-WELDING FITTINGS

Socket-welding fittings 2 inches or smaller in size, such as elbows, tees, and couplings, shall be of forged steel and shall have a working pressure of at least 3000 pounds per square inch.

6.3.4 PACKED-SLEEVE AND RESILIENT-SEALED COUPLINGS

Packed-sleeve and resilient-sealed couplings shall not be used.

6.3.5 FLANGES

Weld-neck flanges are preferred. If slip-on flanges are used, they shall be welded both inside and outside.

6.4 Plugs

Plugs shall be constructed of steel.

6.5 Unions

Unions shall be of forged steel, shall have a working pressure of at least 3000 pounds per square inch, and shall have ground metal-to-metal seats. Gasket unions shall not be used. Unions shall not be used between the vessel and the first valve.

6.6 Valves

6.6.1 PRIMARY SHUTOFF VALVES

6.6.1.1 The primary shutoff valves for a tank (specifically the valves nearest the vessel that can shut off flow) shall be made from steel. Valves constructed of free-machining steel similar to AISI Series 1100 and 1200 shall not be used.

6.6.1.2 Union or screwed-bonnet valves shall not be used unless they are equipped with bonnet retainers or the bonnets are tack welded.

6.6.1.3 Valves that are sandwiched between two flanges by long, exposed bolts shall not be used, unless the valves have lug-type bodies that cover the bolts.

6.6.1.4 Ball valves shall meet the requirements of API Standard 607.

6.6.2 CHECK VALVES

Check valves shall be installed on the discharge side of all centrifugal pumps.

6.6.3 PRESSURE RELIEF VALVES

Pressure relief valves shall be constructed of steel.

6.6.4 THERMAL RELIEF VALVES

Suitable thermal relief valves shall be provided on liquid lines that can be blocked between two shutoff valves. Other equipment that can be blocked between shutoff valves shall be provided with protection from overpressure due to thermal expansion of the liquid. Where liquid is trapped in valve cavities, the need for pressure relief shall be considered.

6.7 Location, Installation, and Flexibility of Piping, Valves, and Fittings

6.7.1 Piping shall be provided with adequate flexibility to accommodate the following:

a. Settling of tanks or shifting of foundations.
b. Expansion or contraction of tanks or piping with changes in temperature.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545859 722 ■

c. Soil movement.
d. Cooling or heating of unloading connections, vent connections, or loading and unloading headers.

6.7.2 Headers located on piers shall be designed to permit unrestrained movement of the piping in the direction of expansion or contraction except at necessary anchor points.

6.7.3 All water drawoffs shall be extended so that they do not terminate under the vessel. Drain lines shall not be directed into a public sewer or into a drain not designed to contain flammable materials. Double valves shall be provided. When drain lines are supported by any type of support not directly attached to the tank, adequate flexibility shall be provided in the lines to accommodate differential settlement. Stress imposed on the vessel nozzle by the drain lines shall be minimized.

6.7.4 Water drain lines and similar small lines shall be adequately supported or shall be fabricated with sufficient strength to be self-supporting under operating conditions, including the condition of maximum flow reaction thrust. Stress imposed on the vessel by the drain lines shall be minimized.

SECTION 7—LOADING, PRODUCT TRANSFER, AND UNLOADING FACILITIES

7.1 Scope

This section covers the design and construction of facilities that transfer LPG as follows:

a. From a pipeline to stationary storage.
b. From truck or railcar racks and marine docks to stationary storage.
c. From stationary storage to truck or railcar racks or marine docks.
d. From stationary storage to a pipeline.

7.2 Rates of Loading and Unloading

7.2.1 SIZING

Pumps and loading devices shall be sized to provide rates of flow appropriate to the capacity of the facility. Care shall be taken to ensure that the rates of flow give the operator enough time to follow the course of loading and unloading at all times and to shut down the facility before tanks are completely emptied or before they are filled beyond their maximum filling height.

7.2.2 DESIGN

The transfer system shall incorporate a means for rapidly and positively stopping the flow in an emergency. Transfer systems shall be designed to prevent dangerous surge pressures when the flow in either direction is stopped.

7.3 Transfer, Loading, and Unloading Equipment

7.3.1 PUMPS

7.3.1.1 Pumps may be centrifugal, reciprocating, gear, submersible or may be another type designed for handling LPG. The design pressure and construction material of the pumps shall be capable of safely withstanding the maximum pressure that could be developed by the product, the transfer equipment, or both. When centrifugal pumps are used, mechanical seals are recommended. Positive displacement pumps shall have a suitable relief device on the discharge side unless other provisions are made for protection of the equipment.

7.3.1.2 When submersible pumps are used, each interface between the LPG system and an electrical conduit or wiring system shall be sealed or isolated to prevent passage of LPG to another portion of the electrical installation. See NFPA 59A for further information.

7.3.2 COMPRESSORS

Compressors for loading and unloading LPG shall be designed for the maximum outlet pressure to which they may be subjected. Each centrifugal compressor discharge connection shall be equipped with a check valve. Each centrifugal compressor shall be evaluated for conditions that may cause overpressure, and a relieving device shall be provided if required. Each positive displacement compressor shall be equipped with a pressure-relieving device on the discharge side. A suitably sized scrubber or liquid knockout drum shall be installed immediately upstream of vapor compressors. The scrubber shall be equipped with a high-liquid-level device to shut down the compressor.

7.3.3 PRESSURE GAUGES

Pressure gauges shall be provided in enough locations in the liquid and vapor lines to enable the operator to monitor operating pressure and pressure differentials constantly to ensure safe operation.

7.3.4 EMERGENCY SHUTOFF VALVES

7.3.4.1 Emergency shutoff valves shall be provided in the loading-unloading system for tank cars, trucks, and marine

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

facilities and shall incorporate the following means of closing:

a. Manual shutoff at the installed location.
b. Manual activation from a location accessible during an emergency.

A safety analysis shall be the basis for determining the need for the following:

a. Automatic shutoff in the event of an LPG release.
b. Automatic shutoff through thermal (fire) actuation.

7.3.4.2 Installation practices for emergency shutoff valves shall include those specified in 7.3.4.2.1 and 7.3.4.2.2.

7.3.4.2.1 When hose or swivel piping is used for liquid or vapor transfer, an emergency shutoff valve shall be installed in the fixed piping of the transfer system within 20 linear feet of pipe from the end to which the hose or swivel piping is connected. Where the flow is in one direction only, a backflow check valve may be used in place of an emergency shutoff valve if the backflow check valve is installed in a dedicated storage vessel fill line or vapor return line. When two or more hoses or swivel piping arrangements are used, either an emergency shutoff valve or a backflow check valve (for unloading lines only) shall be installed in each leg of the piping.

7.3.4.2.2 The emergency shutoff valves or backflow check valves shall be installed in the fixed piping so that any break resulting from a pull will occur on the hose or swivel piping side of the connection while the valves and piping on the plant side of the connection remain intact. This may be accomplished by the use of concrete bulkheads or equivalent anchorage or by the use of a weakness or shear fitting. Refer to NPGA Bulletin 128.

7.3.4.3 Facility boundary limit block valves and check valves shall be provided if the feed or product is transported by pipeline. If block valves are manually operated, they shall be accessible during an emergency.

7.4 Grounding and Bonding

7.4.1 STATIC ELECTRICITY

Protection from discharge of static electricity is not required when a tank car, a tank truck, or marine equipment is loaded or unloaded through tight (top or bottom) outlets using a conductive or nonconductive hose, flexible metallic tubing, or pipe connection because no spark gap exists while product is flowing (see API Recommended Practice 2003).

7.4.2 STRAY CURRENTS

If stray currents are present or if impressed currents are used on the loading and unloading systems for cathodic protection, protective measures shall be taken in accordance with API Recommended Practice 2003.

7.4.3 LIGHTNING PROTECTION

Aboveground metallic LPG storage containers do not require lightning protection. To protect personnel and foundations where the piping might not provide grounding, grounding rods shall be provided for tanks supported on nonconductive foundations. See API Recommended Practice 2003 for additional information on lightning protection.

7.5 Hose and Other Flexible Connectors for Product Transfer

7.5.1 HOSE

7.5.1.1 Hose shall be fabricated of materials resistant to LPG in both liquid and vapor form. If wire braid is used for reinforcement, it shall be made from corrosion-resistant material such as stainless steel.

7.5.1.2 The correctness of design, construction, and performance of hose shall be determined. Only hose listed by Underwriters Laboratories or another nationally recognized testing laboratory shall be used for LPG transfer applications. Hose used in marine applications shall be approved by the U.S. Coast Guard.

7.5.1.3 Hose, hose connections, and flexible connectors used for transferring LPG liquid or vapor at pressures in excess of 5 pounds per square inch gauge shall conform to the criteria specified in 7.5.1.3.1 through 7.5.1.3.3.

7.5.1.3.1 Hose shall be designed for a minimum working pressure of 350 pounds per square inch gauge and a minimum bursting pressure of 1750 pounds per square inch gauge. Hose shall be marked "LPG" or "LP-gas" at intervals of not more than 10 feet.

7.5.1.3.2 After the installation of connections, hose assemblies shall be capable of withstanding a test pressure of not less than 700 pounds per square inch gauge.

7.5.1.3.3 Hose assemblies shall be visually inspected before each use for damage or defects. Hose assemblies shall be tested at least annually at whichever is greater, the maximum pump discharge pressure or the relief valve setting.

7.5.2 HOSE PROTECTION

Hose shall be protected from the elements and physical damage. Particular attention shall be given to the prevention of potentially damaging ice formation on the corrugations of metallic hose.

7.5.3 SUPPORT OF LOADING ARMS OR HOSES

Provisions shall be made for adequately supporting the loading hose or arm. The weight of ice formations on uninsulated hoses or arms shall be considered in the design of counterweights.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545861 380 ■

7.5.4 FLEXIBLE PIPE CONNECTION

Each flexible pipe connection shall be capable of withstanding a test pressure of 1½ times the design pressure for its part of the system.

7.6 Blowdown or Venting of Loading and Unloading Lines

Each hose or pipe connection(s) with flexible joints used in the loading and unloading of LPG between stationary and mobile tanks shall be equipped with a blowdown or bleeder valve. The valve shall enable the emptying of the hose or pipe connection(s) after the block valves on each side of the hose or pipe connection(s) have been closed. The blowdown or bleeder valve shall be sized and installed so that venting does not create a hazard.

7.7 Marking of Valves in Loading and Unloading Systems

When more than one product is handled at a loading or unloading rack, the lines shall be marked or designated so that the operator can identify the various lines and valves without having to trace them to their source or destination.

7.8 Metering Equipment Used in Loading and Unloading

When liquid meters are used to measure the volume of LPG that is being transferred from one container to another or that is being transferred to or from a pipeline, the meters and accessory equipment shall be installed in accordance with the procedures stipulated by API Recommended Practice 550 (out of print), API Recommended Practice 551, and Chapter 5 of the API *Manual of Petroleum Measurement Standards*.

7.9 LPG Odorization

If specified, a stationary LPG storage facility designed to transfer LPG to tanks, trucks, railroad tank-cars, or marine containers through loading racks or docks shall have equipment that enables the addition of odorant as specified by NFPA 58 and the Transportation Safety Act of 1974, Part 173, Section 315.

SECTION 8—FIRE PROTECTION

8.1 General

Fire protection provisions shall be based on a safety analysis of local conditions, exposure from or to other sites, availability of a water supply, and effectiveness of fire brigades and fire departments. The analysis shall include possible but realistic accident scenarios that may occur, including scenarios of vapor release, ignition, and fire. For additional information, background, and guidance, see API Publication 2510A.

8.2 Access for Fire Fighting

The layout of the storage facility, including the arrangement and location of plant roads, walkways, doors, and operating equipment, shall be designed to permit personnel and equipment to reach any area affected by fire rapidly and effectively. The layout shall permit access from at least two directions. Emergency escape as well as access for fire fighting shall be considered.

8.3 Fire Water Use

Storage facilities for LPG shall be provided with a fire water system unless a safety analysis shows this protection to be unnecessary or impractical. See API Publication 2510A for additional information.

8.3.1 SYSTEM DESIGN

The design of the fire water system shall be in accordance with 8.3.1.1 through 8.3.1.10.

8.3.1.1 A looped fire water system shall be provided around the storage and handling portions of an LPG facility.

8.3.1.2 Sufficient isolation valves shall be provided in the fire water grid to prevent loss of the grid due to a single break in the water main. Block valves shall be arranged so that all parts of the plant can be protected by a portion of the fire water main system when an impaired section is isolated for repair.

8.3.1.3 The capacity of the fire water system shall be equal to the amount of fire water required to cool the largest vessel being protected, plus the amount required to cool adjacent vessels, plus reserve capacity for up to three additional 250-gallon-per-minute cooling streams. Where the capacity of the fire water system is determined by the requirement for LPG storage, the system is permitted to be sectionalized to reduce the maximum simultaneous requirement for fire water.

8.3.1.4 Pipe used for fire water mains and branch lines to hydrants shall be at least 6 NPS in size. Branch lines to deluge, monitor, or spray systems are permitted to be smaller, provided hydraulic calculations show that the size selected will supply the design demand at the required pressure.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

8.3.1.5 The fire water system shall be functional in all seasons and shall be capable of delivering 100 percent of the design rate for at least 4 hours. The fire water system shall be suitably protected from freezing where necessary.

8.3.1.6 The fire water grid shall be designed so that at least half the water required by the single largest incident can be delivered if any single section of the fire water main is lost.

8.3.1.7 Regardless of the fire water application method used, the location of hydrants shall be arranged so that each storage vessel can be reached from at least two directions by at least three cooling streams none of which uses more than 300 feet of hose.

8.3.1.8 The fire water system shall be designed to provide water to the protected equipment within 60 seconds of activation to achieve design water delivery rates within 10 minutes of system activation.

8.3.1.9 The fire water system shall be designed to facilitate testing to assure reliability, adequate flow rate, and adequate coverage of the protected equipment.

8.3.1.10 The fire water systems shall be tested to verify that their performance is as designed. Since the capacity of the water grid can deteriorate gradually as a result of scale buildup in the water mains, a Hazen-Williams coefficient no greater than 100 shall be used for unlined steel pipe.

8.3.2 FIRE WATER APPLICATION METHODS

LPG storage vessels shall be protected by water deluge systems, fixed monitors, water spray systems, or any combination of these systems. Portable equipment may be used but shall not be a primary method of water application.

8.3.2.1 Water Deluge System

A water deluge system is a system in which all the water is applied at the top of the vessel and allowed to run down the sides. When a water deluge system is selected for the protection of LPG storage facilities, it shall include the design features described in 8.3.2.1.1 through 8.3.2.1.5.

8.3.2.1.1 The system shall be designed so that under nonfire conditions, the water flows evenly over the entire surface of the vessel. The adequacy of the water coverage shall be determined by means of performance tests.

8.3.2.1.2 If weirs are used to improve distribution, they shall be provided with drainage to prevent standing water, which may increase corrosion.

8.3.2.1.3 Pipe used for main water distribution lines shall have a diameter of at least 3 inches.

8.3.2.1.4 Top-mounted water distribution nozzles shall be at least 1½ inches in size and shall be provided with suitable deflectors or weirs to achieve good water distribution.

8.3.2.1.5 The system shall be manually operated from a safe location that is outside the spill containment area and that is at least 50 feet from the vessel being protected. The location of the actuating valve shall be clearly and prominently marked. In locations with unattended or partially attended operations, consideration shall be given to additional methods of system activation such as automatic or remote operation. When the system is remotely or automatically operated, a full-size manually operated bypass valve shall also be provided in an accessible, safe location.

8.3.2.2 Fixed Monitors

Fire water monitors permanently connected to the fire water grid can be used to apply cooling water to the shell of LPG storage vessels. Where protection by means of monitors is selected, the system shall include the design features described in 8.3.2.2.1 through 8.3.2.2.4.

8.3.2.2.1 The entire surface of each vessel shall be reached with streams from the monitors.

8.3.2.2.2 Each monitor shall be accessible during a fire or shall be remotely activated and controlled.

8.3.2.2.3 Monitor nozzles shall be adjustable for fog or straight stream, as required, to provide the most effective coverage of the protected vessel.

8.3.2.2.4 In freezing climates, monitors shall be suitably protected against freezing.

8.3.2.3 Water Spray Systems

A water spray system uses many spray nozzles arranged in a grid pattern to distribute the water evenly over the LPG vessel. When a water spray system is selected for the protection of LPG storage facilities, it shall include the design features described in 8.3.2.3.1 through 8.3.2.3.6.

8.3.2.3.1 The system shall be designed so that the water is applied evenly over the entire surface of the vessel that may be exposed to fire. Allowance for rundown is permitted. The adequacy of the water coverage shall be determined by performance tests.

8.3.2.3.2 The spray system shall be an open-head system, with all nozzles supplied from the top of the supply branch line and each branch line shall be from the top of the water distribution main line. Spray orifice size shall be at least 0.25 inch. Larger orifice sizes will reduce the tendency of the nozzles to become clogged.

8.3.2.3.3 The system shall be manually operated from a safe location that is outside the spill containment area and that is at least 50 feet from the vessel being protected. The location of the actuating valve shall be clearly and prominently marked. In locations with unattended or partially

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545863 153 ■

attended operations, consideration shall be given to additional methods of system activation such as automatic or remote operation. When the system is remotely or automatically operated, a full-size manually operated bypass valve shall also be provided in an accessible, safe location.

8.3.2.3.4 Flush-out connections shall be installed in the system to permit flushing at periodic intervals. Accessible low-point drain connections shall also be provided.

8.3.2.3.5 The sizing of all piping shall be based on hydraulic calculations. Pipe used for main water distribution lines shall have a diameter of at least 3 inches. Pipe used for branch lines to spray heads is permitted to not be less than ¾ NPS in size.

8.3.2.3.6 A full-flow strainer with a valved blow-off connection shall be installed in the main feeder line to the spray system. The maximum size of the opening in the strainer shall be 0.25 inch. A full-size valved bypass shall be provided.

8.3.2.4 Portable Equipment

Portable equipment, such as fire hoses and portable monitors, shall not be used as the only means of protecting exposed LPG vessels. It is permitted to use portable equipment when vessels are fireproofed as outlined in 8.7.

8.3.3 FIRE WATER APPLICATION RATES

8.3.3.1 The minimum required fire water application rate depends on the method of application.

8.3.3.2 In determining fire water application rates, the surface area of the vessel that could be exposed to fire shall be the surface area of the vessel above the level of the liquid contents at the vessel's lowest operating level.

8.3.3.3 Fixed deluge or water spray systems shall be designed to protect against pool fire exposure to the vessel with a minimum fire water application rate of 0.10 gallon per minute per square foot of exposed vessel surface. To cool a vessel receiving high-intensity flame impingement and to protect against vessel failure, an application rate of at least 0.25 gallon per minute per square foot shall be used in the design.

8.3.3.4 To compensate for losses due to wind and vaporization that occur before the stream reaches the vessel wall, fire water monitor systems shall be designed to protect against pool fire exposure to the vessel with a minimum water application rate of 0.20 gallon per minute per square foot of exposed vessel surface.

8.4 Fire Detection Systems

A safety analysis shall be used to determine the need for fire and hydrocarbon detection systems. Where provided, fire and hydrocarbon detection systems shall be arranged to sound their alarms whenever fire or hydrocarbons are present. It is permitted to use detection systems to automatically activate isolation or fire protection systems in remote or unattended facilities.

8.5 Fire Extinguishers

8.5.1 Portable fire extinguishers shall be used to extinguish an LPG fire only after the source of LPG has been shut off, to prevent the formation of a hazardous vapor cloud.

8.5.2 Dry chemical fire extinguishers shall be provided at strategic locations such as those near pumps and loading racks so that they are readily available for operator use.

8.6 Fire-Fighting Foam

Fire-fighting foam shall not be used to extinguish LPG fires.

8.7 Fireproofing of LPG Vessels

8.7.1 Except for remote facilities, which require no protection, fireproofing shall be used to protect vessels if portable equipment is the only means of applying fire water.

8.7.2 Where fireproofing is used, it shall provide protection of the structural steel or LPG vessel for the time period required for operation of fire water systems.

8.7.3 When fireproofing is used, it shall comply with the provisions of 8.7.3.1 through 8.7.3.5.

8.7.3.1 Outside surfaces of LPG vessels that may be exposed to fire shall be covered with a fireproofing material that is suitable for the temperatures to which the vessel will be exposed. Refer to API Publication 2218 for additional information on fireproofing.

8.7.3.2 The thickness of the fireproofing material should be equivalent to a fire endurance of 1½ hours per UL 1709 when tested on a 10W49 column.

8.7.3.3 Thermal insulation used for fireproofing shall be jacketed with rust-resistant steel.

8.7.3.4 The fireproofing material shall be suitably protected against weather damage and sealed to prevent water entry.

8.7.3.5 The fireproofing system shall be capable of withstanding exposure to direct flame impingement and shall be resistant to dislodgment by direct impingement of fire water streams. Refer to NFPA 58, Appendix G, for further information.

8.8 Fireproofing of Structural Supports

8.8.1 Except for remote facilities, which require no protection, structural supports shall be provided with fireproofing, as specified in 8.8.2 through 8.8.9.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545864 09T ■

8.8.2 Fireproofing shall be provided on the aboveground portions of the vessel's supporting structures. The fireproofing shall cover all support members required to support the static load of the full vessel. Fireproofing shall not encase the points at which the supports are welded to the vessel. Refer to API Publication 2218 for additional information on fireproofing.

8.8.3 Fireproofing shall be provided on horizontal vessel saddles where the distance between the bottom of the vessel and the top of the support structure is greater than 12 inches. Where such fireproofing is provided, it shall extend from the support structure to the vessel, except that it shall not encase the points at which the saddles are welded to the vessel.

8.8.4 When a vertical vessel is supported by a skirt, the exterior of the skirt shall be fireproofed.

8.8.5 Fireproofing shall be provided on all pipe supports within 50 feet of the vessel and on all pipe supports within the spill containment area of the vessel.

8.8.6 To be considered as adequately fireproofed, support structures of concrete or masonry shall meet the criteria of 8.8.8.

8.8.7 Fireproofing is not required for diagonal bracing, including tie rods, or for redundant members that are not necessary for supporting static loads.

8.8.8 The thickness of the fireproofing material should be equivalent to a fire endurance of 1½ hours per UL 1709 when tested on a 10W49 column.

8.8.9 Fireproofing material shall be suitably protected against weather damage and sealed to prevent water entry. It shall be resistant to dislodgment by direct impingement of fire water streams.

8.9 Burying and Mounding

CAUTION: LPG vessels buried below grade or mounded above grade to reduce exposure to an external fire require special precautions, careful preparation, and special design features. Adequate protection against corrosion, leaks, and mechanical damage when the vessel is uncovered for inspection shall be provided. Burying and mounding for protection of LPG storage vessels shall be specially engineered and arranged to meet the provisions of NFPA 58 for buried or mounded tanks.

8.10 Electrical Installations and Equipment

All electrical installations and equipment shall conform to the provisions of NFPA 70. Refer to API Recommended Practice 500 for guidance in the classification of electrical areas.

8.11 Critical Wiring and Control Systems

8.11.1 Unless the electrical, instrument, and control systems are fail-safe in a fire, these systems—including especially the wiring used to activate the equipment needed in an emergency—shall be protected from fire damage. Thus, in areas where the control wiring used to activate an emergency shutoff valve during a fire could be exposed to the fire, the wiring shall be protected against a 15-minute fire exposure; however, if activation of an emergency shutoff valve would not be necessary during any fire to which its wiring could be exposed, then protection of the wiring is not required.

8.11.2 Wiring shall be protected by selective routing, burying, fireproofing, or a combination of these methods.

8.12 Safety Precaution Signs

Appropriate safety precaution signs shall be placed to provide notification and instructions concerning safety requirements and emergency systems.

8.13 Lighting

In all storage and operating areas, lighting that is adequate for operations under normal conditions shall be provided. In addition, lighting that is sufficient to enable safe operations during an emergency shall be provided.

8.14 Fencing

Any LPG storage installation that is not within a fenced plant area or otherwise isolated from the public shall be fenced, and at least two means of exit shall be provided. Exits shall be located so that a single emergency cannot prevent egress from any part of the installation.

8.15 Roadways

Suitable roadways or other means of access for fire-fighting equipment such as wheeled extinguishers or fire trucks shall be provided. Access to LPG handling and storage areas shall be restricted or controlled.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545865 T26 ■

SECTION 9—REFRIGERATED STORAGE

9.1 General

9.1.1 SCOPE

This section contains specific requirements for refrigerated LPG tanks. Also, unless specifically superseded or expanded upon in this section, the requirements of previous sections apply to refrigerated storage.

9.1.2 ROLLOVER PROTECTION

Temperature-density gradients may develop in refrigerated storage tanks during loading if the mixing between the existing tank contents and the material being loaded into the tank is insufficient. The potential for stratification shall be reduced by the use of an inlet nozzle designed to promote mixing or by the use of mechanical agitation. Regardless of the mechanism selected, it shall be designed to provide sufficient energy to adequately mix the tank contents.

9.2 Design Requirements

9.2.1 CODE REQUIREMENTS

9.2.1.1 Low-Pressure Tanks

Tanks with design pressures of less than 15 pounds per square inch gauge shall conform to API Standard 620.

9.2.1.2 Pressure Storage

Tanks with design pressures of at least 15 pounds per square inch gauge shall be designed in accordance with the ASME *Boiler and Pressure Vessel Code*, Section VIII, Division 1 or 2.

9.2.2 DESIGN PRESSURE

9.2.2.1 The design pressure of a refrigerated LPG tank is determined by the product's vapor pressure at the storage temperature. The set pressure of the pressure-relieving device shall be at least 5 percent greater than the design operating pressure.

9.2.2.2 The tank section above the maximum liquid level shall be designed for a pressure of at least that at which the pressure relief valves are to be set and for the maximum partial vacuum that can be developed. All portions of the tank below the maximum liquid level shall be designed for at least the most severe combination of gas pressure (or partial vacuum) and static liquid head affecting each element of the tank.

9.2.3 DESIGN TEMPERATURE

The design temperature for a refrigerated LPG tank shall be the lowest of the following:

a. The lowest temperature to which the tank contents will be refrigerated.
b. The lowest shell temperature resulting from cold ambient conditions, if that temperature is below the refrigerated product temperature.
c. The autorefrigeration temperature of the contents.

9.3 Siting Requirements

9.3.1 MINIMUM DISTANCE REQUIREMENTS FOR REFRIGERATED LPG TANKS

9.3.1.1 The minimum horizontal distance between the shell of a refrigerated LPG tank and the line of adjoining property that may be developed shall be 200 feet. Where residences, public buildings, places of assembly, or industrial sites are located on adjacent property, greater distances or other supplemental protection shall be evaluated.

9.3.1.2 The minimum horizontal distance between the shells of adjacent refrigerated LPG tanks shall be half the diameter of the larger tank.

9.3.1.3 The minimum horizontal distance between the shell of a refrigerated LPG tank and the shell of another nonrefrigerated hydrocarbon storage facility shall be the largest of the following distances with the exception noted after Item d:

a. If the other storage is pressurized, three quarters of the larger tank diameter.
b. If the other storage is in atmospheric tanks and is designed to contain material with a flash point of 100°F or less, one diameter of the larger tank.
c. If the other storage is in atmospheric tanks and is designed to contain material with a flash point greater than 100°F, half the diameter of the larger tank.
d. 100 feet.

The minimum horizontal distance between shells need not exceed 200 feet.

9.3.2 SITING OF REFRIGERATED LPG TANKS

Refrigerated LPG tanks shall not be located within buildings, within the spill containment areas of other flammable or combustible liquid storage tanks as defined in NFPA 30, or within the spill containment areas of pressurized storage tanks.

9.3.3 SPILL CONTAINMENT

9.3.3.1 Refrigerated LPG tanks shall be provided with spill containment facilities. To prevent the accumulation of flammable material under or near a refrigerated LPG tank,

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545866 962 ■

the ground under and surrounding the tank shall be graded to drain any spills to a safe area away from the tank.

9.3.3.2 Spill containment shall be provided by the remote impoundment of spilled material or by the diking of the area surrounding the vessel.

9.3.4 REMOTE IMPOUNDMENT

9.3.4.1 If remote impoundment is to be used for spill containment, the remote impoundment facility shall be designed according to the guidelines given in 9.3.4.2 through 9.3.4.5.

9.3.4.2 The grading of the area under and surrounding the vessels shall direct any leaks or spills to the remote impoundment area. The grading shall be a minimum of 1 percent slope.

9.3.4.3 Toe walls, dikes, trenches, or channels may be used to assist in draining the spilled product from the area of the tank to a remote impoundment area. However, the use of trenches or channels shall be minimized.

9.3.4.4 The remote impoundment area shall be located at least 50 feet from the vessels draining to it and from any piping or other equipment.

9.3.4.5 The holdup of the remote impoundment area shall be at least 100 percent of the volume of the largest vessel draining to it.

9.3.5 DIKING

9.3.5.1 If diking around the vessel is to be used for spill containment, the diked area shall be designed according to the guidelines given in 9.3.5.2 through 9.3.5.4.

9.3.5.2 The grading of the area under and surrounding the vessel shall direct any leaks or spills to the edge of the diked area. The grading shall be a minimum of 1 percent slope. Within the diked area, the grading shall cause spills to accumulate away from the vessel and any piping located within the diked area.

9.3.5.3 Each refrigerated LPG tank shall be provided with its own diked area. The holdup of the diked area shall be at least 100 percent of the volume of the tank.

EXCEPTION: Two tanks of the same basic design may be enclosed within the same diked area as long as an intermediate dike is provided between the tanks. The intermediate dike shall be 1 foot lower than the peripheral dike. The holdup of the diked area shall be at least 100 percent of the volume of the larger vessel within it after allowing for the displacement of the other vessel or tank within the same enclosure.

9.3.5.4 When dikes are used as part of the spill containment system, the minimum height shall be 1.5 feet, measured from the inside of the diked area. Where dikes must be higher than 6 feet, provisions shall be made for normal and emergency access into and out of the diked enclosure. Where dikes must be higher than 12 feet or where ventilation is restricted by the dike, provision shall be made for normal operation of valves and access to the top of the tank or tanks without the need for personnel to enter into the area of the diked enclosure that is below the top of the dike. All earthen dikes shall have a flat top section at least 2 feet wide.

9.4 Thermal Considerations

The tank foundation shall be designed to prevent 32°F or lower temperatures from penetrating the pad and soil. This limitation shall be accomplished by ventilation, insulation, heating systems, or a combination of these. Heating elements, controls, and temperature sensors shall be designed for easy access and replacement while the tank is in service. Foundation heating systems shall be provided with temperature monitoring and controls. The design of the supporting structure shall consider loads resulting from (a) the thermal gradient across the supporting structure, foundation, and piling due to the temperature of the contents of the vessel and (b) the thermal shock from accidental spills.

9.5 Tank Accessories

9.5.1 PRESSURE/VACUUM-RELIEVING DEVICES

9.5.1.1 Each refrigerated LPG tank shall be provided with at least one pressure-relieving device set to discharge at no more than the maximum allowable working pressure of the tank.

9.5.1.2 Tanks that may be damaged by internal vacuum shall be provided with at least one vacuum-relieving device set to open at not less than the partial vacuum design pressure.

9.5.1.3 When a closed inner-tank design is used with an outer vapor-tight shell, the outer shell shall be equipped with one or more pressure/vacuum-relieving devices.

9.5.2 RELIEF VALVE CAPACITIES

Relief devices for tanks designed to conform to API Standard 620 shall be designed in accordance with API Standard 2000. Relief devices for tanks designed to conform to Section VIII of the ASME Code shall be designed in accordance with API Recommended Practice 520.

9.5.3 TEMPERATURE INDICATORS

Each tank shall be fitted with thermocouples or equivalent temperature-indicating devices for use during cooldown and operations.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545867 8T9 ■

9.5.4 SAMPLING CONNECTIONS

If sampling connections are required, they shall be installed on the tank piping rather than on the tank.

9.5.5 TANK ACCESSORY MATERIALS

Low-ductility material such as cast iron, semisteel, malleable iron, and cast aluminum shall not be used in any pressure-retaining accessory parts.

9.6 Piping Requirements

9.6.1 VALVES

Shutoff valves and accessory equipment shall be constructed of material suitable for the operating pressure and temperature extremes to which they may be subjected.

9.6.2 INSULATION

The insulation shall comprise or contain a vapor barrier and shall be weatherproofed. Insulation and weatherproofing shall be fire retardant. Steel surfaces covered by insulation shall be properly coated to prevent corrosion.

9.6.3 LOCATION

When cold piping is routed below grade, trenches, casing, or other means shall be used to permit expansion and contraction of the piping.

9.6.4 MULTIPLE PRODUCT TYPES

When a storage facility handles more than one type of product, dedicated loading and unloading lines between tanks and racks shall be considered for each type of product.

9.7 Refrigeration System

9.7.1 LPG TEMPERATURE

The refrigeration system shall maintain the LPG at a temperature at which the LPG's vapor pressure does not exceed the tank's design pressure.

9.7.2 SIZING

The sizing of the refrigeration system shall consider the following factors:

a. Heat flow from the following sources:
   1. The difference between the design ambient temperature and the design storage temperature.
   2. Maximum solar radiation.
   3. Receipt of product that is warmer than the design temperature, if such an operation is expected.
   4. Foundation heaters.
   5. Connected piping.

b. Vapor displacement during filling and vapor return during product transfer.

9.7.3 VAPOR HANDLING

An alternate handling method shall be provided for an excess in the LPG vapor load resulting from insufficient refrigeration or loss of refrigeration.

9.7.4 PRESSURE-RELIEVING DEVICES

Refer to API Recommended Practice 520, Parts I and II, for the proper design of pressure-relieving devices and systems for process equipment used in liquefaction and vaporization facilities.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545868 735 ■

APPENDIX A—PIPING, VALVES, FITTINGS, AND OPTIONAL EQUIPMENT

A.1 Optional Equipment

A.1.1 GENERAL

Tanks may be fitted with the optional equipment described in A.1.2 through A.1.7. Any optional equipment selected for use shall be suitable for use with LPG and designed for at least the maximum service conditions to which it may be subjected.

A.1.2 SAMPLING CONNECTIONS

Sampling connections may be provided on tanks. As an alternative, the connections on gauging equipment may be used for sampling if they are suitably located. Adequate bracing of small connections and piping in sampling lines shall be provided to minimize vulnerability to mechanical damage. The inlet piping to sample containers shall be double valved. Sample connection locations should not be under the vessel. Connections shall be oriented so that purge vapors do not engulf the operator or approach an ignition source.

A.1.3 AUTOMATIC AND REMOTE DEVICES

Automatic shutoff valves, remotely operated shutoff valves, automatic warning devices, pump shutdown switches, or a combination of these may be used where tanks are operated remotely, where they receive LPG at a high rate of flow, or for other circumstances in which the designer considers it advisable. Fireproofing of the control systems may be required for these devices to be effective during fire exposure. See 8.11 for additional information.

A.1.4 STAIRS, LADDERS, WALKWAYS, AND PLATFORMS

Suitable stairs, ladders, walkways, and platforms should be provided to allow access to operating valves and equipment.

A.1.5 COMMON DISCHARGE HEADER

Pressure relief valve lines for one or more tanks may be connected to a common discharge header provided the LPG is being discharged to a flare. Common discharge headers shall be designed in accordance with API Recommended Practice 521 and shall comply with the provisions of 5.1.6.5. Back pressures that could develop during relief valve discharge shall be taken into account when determining the size of the relief device and of the discharge header. For pilot-operated relief valves discharging into a common header, the effects of backflow should be considered, and a backflow preventer should be provided if required.

CAUTION: Common headers should not be used for venting to the atmosphere. Common discharge headers shall be sized for full relief capacity of all tanks that could be involved in a single emergency situation. Liquid traps in the common header shall be prevented. Other vents, drains, bleeders, and pressure relief devices shall not be tied into the common discharge header if back pressures can develop that may prevent proper functioning of the pressure relief devices on the tank. See API Publication 2510A for additional information.

A.1.6 WATER DRAWOFFS

Facilities for removing water from LPG storage vessels should be provided. These water drawoffs shall be designed to prevent freezing of water within them. See 6.7.4, 6.7.5, and API Publication 2510A for additional information.

A.1.7 WATER FLOOD CONNECTION

Each LPG storage vessel may be provided with a water flood connection. The water flood connection may be provided either into the vapor space of the vessel or directly into the product line to the bottom of the vessel. When the water flood connection is provided on the product line, the possibility of water freezing shall be considered in the design. The water flood connection shall extend outside the spill containment system and shall include (in physical order) a block valve, a check valve, and provision for connection to the water system.

A.2 Location, Installation, and Flexibility of Piping, Valves, and Fittings

A.2.1 RECOMMENDED PRACTICES

A.2.1.1 The practices described in A.2.1.2 through A.2.1.11 concerning location, installation, and flexibility of piping, valves, and fittings are recommended.

A.2.1.2 The design of header piping and tank loading and unloading connections should be as simple as possible. The number of connections to the storage vessel should be minimized. Operating errors increase as the complexity of the piping installation and the number of connections increase.

A.2.1.3 Shutoff valves that must be used during normal operations should be accessible to the operator and should be as close to the tanks, pumps, compressors, and other compo-

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

API STD*2510 95 ■ 0732290 0545869 671 ■

nents as practical. This recommendation should not be construed as discouraging installation of remotely operated shutoff valves or other safety devices.

A.2.1.4 Headers may be installed on piers, supported by stanchions, or buried, as specified by the owner. Buried lines have the advantage of being protected from fires and explosions but have the disadvantages of the possibility of soil corrosion, inaccessibility for inspection, and reduction in flexibility because of the binding action of the earth. Buried lines should be installed below the frost line and protected from corrosion.

A.2.1.5 Piping should not be laid under concrete floors or slabs. When piping must extend through a concrete wall or below a floor slab, it should be protected by a suitable casing.

A.2.1.6 Lines laid under railroad tracks, highways, access roads, or loading slabs should be installed in accordance with API Recommended Practice 1102.

A.2.1.7 Interconnected piping between tanks or tank accessories should be installed to permit flexibility in all planes. For example, loading and unloading headers should not be connected to a tank by short, straight rigid piping, regardless of whether the piping is screwed or welded. Equalizing piping should not be connected by short, straight piping between tanks. Vent or relief piping should not have straight piping between adjacent tanks. Piping should include adequate lengths of pipe, with changes in direction obtained by the use of elbows or bends, to provide for possible vertical and horizontal movement of the header relative to the tank.

A.2.1.8 In piping where thermal expansion and contraction are expected to occur, each line should be designed with an adequate expansion bend, angular offset, or other provision to allow for linear movement. Expansion bends may be fabricated from straight lengths of pipe and welding elbows or U-bends. Suitable bellows-type expansion joints, properly anchored and guided, should be used only where space limitations prevent installation of loops or bends.

A.2.1.9 To minimize the amount of material that can be spilled in the event of a line or equipment failure, emergency shutoff valves should be installed in long runs of piping that are used to carry liquids.

A.2.1.10 Low points in piping in which water can accumulate should be avoided to the greatest extent practical. In freezing climates, appropriate freeze protection should be provided where low points cannot be avoided.

A.2.1.11 The second valve in a water drain line should be self-closing (that is, it should be a deadman valve).

A.2.2 REFRIGERATION SYSTEM

A.2.2.1 The vapor load resulting from refrigeration may be handled by one or a combination of the following methods:

a. Recovery by a liquefaction system.
b. Use as a fuel.
c. Use as process feedstock.
d. Disposal by flaring or another safe method.

Alternative handling methods shall be provided to dispose of vented vapors in case of failure of the normal methods. If compressors are used, castings shall be designed to withstand a suction pressure of at least 121 percent of the tank design pressure.

A.2.2.2 A refrigerated LPG system should incorporate the following accessories:

a. An entrainment separator in the compressor suction line.
b. An oil separator in the compressor discharge line (unless the compressor is a dry type).
c. A drain and a gauging device for each separator.
d. A noncondensable gas purge for the condenser.
e. Automatic compressor controls and emergency alarms to signal at the following times:
   1. When any tank's pressure approaches the maximum or minimum allowable tank working pressure or the pressure at which the vacuum vent will open, or
   2. When excess pressure builds up at the condenser because of a failure of the cooling medium.

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

1-01400—5/95—1.1M (1E)

COPYRIGHT 2000 American Petroleum Institute Information Handling Services,
June 05, 2000 01:41:10

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005



Order No. 822-25107

COPYRIGHT 2000 American Petroleum Institute
June 05, 2000 01:41:10

Information Handling Services,